cy audio amplifier/driver, etc. Here we will try to make a fixed-frequency sine wave oscillator that produces four frequencies.

IC LM272 is a low-cost, easy-to-use, unity-gain, stable, dual-power operational amplifier (op-amp) that is mainly used in control applications like servo motors, power supplies, compact discs, VCRs, etc. But LM272M used in this circuit can do much more than that.

Circuit and working

Circuit diagram of the sinusoidal signal generator with forward and inverted outputs, using power amplifier LM272M, is shown in Fig. 1. Besides IC LM272M (IC1), the circuit makes use of +9V voltage regulator LM78M09 (IC2), -9V voltage regulator LM79M09 (IC3), 1N4007 rectifier diodes (D1-D8), and a few other components.

IC LM272M can produce output current up to 1A and can operate from ±2V to ±14V. Its output has low saturation voltage of around 1.5V at output current of 0.5A. The gain product bandwidth is 350kHz and the typical slew rate is 1V/µs. The typical distortion is around 0.5%. This IC has many more applications.

The circuit uses Wien bridge oscillator built around IC1A and voltage regulators 78M09 and 79M09 for producing ±9V to operate the circuit. The table on first page indicates how to select the frequencies.

To calculate the frequency, you can use the classical formula for the Wien bridge oscillator:

$F = 1/(6.28 \times R \times C)$

The resistors and the capacitors should preferably have ±2% or better tolerance. The output amplitude of IC1A is set with the trimmer potmeter VR1 (VR1A1 and VR1B1). You can replace the trimmer potmeter VR1 with appropriate fixed resistors, depending on the electrical bulb LA1.

The circuit works with ±12V regulated power supply. The +12V supply is given to +VIN1 terminal

PARTS LIST

Semiconductors:	
IC1 (IC1A-IC1C)	- LM272M op-amp
IC2	- +9V voltage regulator, LM78M09
IC3	- -9V voltage regulator, LM79M09
D1-D8	- 1N4007 rectifier diode
Resistors (all 1/4-watt, ±2% tolerance):	
R1, R2, R4	- 1-kilo-ohm
R2	- 47-ohm
R5, R6	- 2-kilo-ohm
R7, R10	- 1.8-kilo-ohm
R8, R11	- 180-ohm
R9, R12	- 20-ohm
R13, R14	- 10-ohm
VR1 (VR1A1, VR1B1)	- 470-ohm stereo potmeter
VR2 (VR2A1, VR2B1)	- 10-kilo-ohm stereo potmeter
VR3	- 330-ohm potmeter
VR4-VR5	- 470-ohm potmeter
Capacitors (preferably with 2% or less tolerance):	
C1, C4	- 10nF ceramic disc
C2, C5, C7, C9, C11, C13, C14, C16	- 100nF ceramic disc
C3, C6	- 1µF polyester
C8, C12	- 1000µF, 25V electrolytic
C10, C15	- 220µF, 25V electrolytic
Miscellaneous:	
S1 (S1A1, S1B1), S2 (S2A1, S2B1), S3 (S3A1, S3B1)	- Stereo on/off switch
LA1	- 6V, 20mA/6V, 40mA/12V, 20mA
J1, J4	- 3-pin terminal connector
J2, J3	- 4-pin terminal connector
	- Heat-sink for TO220 package (2 pcs)

and the -12V supply to -VIN1 terminal. In place of +12V and -12V, you can connect +9V and -9V across +VIN2 and -VIN2, respectively, if ±12V is not available. The circuit can work with up to ±14V, but it is suggested to keep the power supply of the IC around ±9V, because the IC can overheat if it is loaded with high current.

The circuit has two outputs, one of which is normal and available across terminal J2, and the other is inverted and available across terminal J3. The first output OUT1 gives a signal with frequencies produced from the first oscillator IC1A. The amplitude of these outputs is adjusted with potmeter VR4. OUT1:10 is divided by 10 of OUT1, while OUT1:100 is divided by 100 of OUT1.

The inverted output gives signal with frequencies produced from